MEHMET ALI ŞAHIN

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Turkey / Ankara

05 January, 1986

PROFILE

I am a software engineer with 10+ years of experience in embedded software and graphical user interface (GUI) development in defence industry since 2010. As a software engineer, it is my greatest desire to improve myself further.

I am experienced in C, C++ & C# programming languages in many software development environment. I have experienced in embedded system software with multi core DSPs & MCUs, Software Development Life Cycle (SDLC), object oriented programming (OOP), Real Time Operating Systems (RTOS), Unified Modelling Language (UML), design patterns, state machines, data structures, multithreading & asynchronous programming, compiling & linking processes, device drivers, APIs, HAL, IRQs & ISRs, using debugging tools in many projects. I also have embedded linux experience in different projects. I am actively using software version tracking tools like Tortoise SVN & Git. As a remote repository hosting service, I used Bitbucket & Gitlab. For Agile sw development, I am actively using Atlassian tools like Jira, Jenkins & Confluence. I am also experienced in writing SRS & SDD documents.

I am familiar with many CPU / MCU / soft-CPU architecture like TI C6000 DSP family, TI C2000 Delfino MCU family, Microsemi SmartFusion2, NXP i.MXRT1050 MCU (Arm CortexM7), TI Tiva MCU (Arm CortexM4).

Although I am an embedded software engineer, I have a lot of experience in GUI development with VisualStudio(C#), QT(C++), LabwindowsCVI(C) since I also develop simulators & test GUIs. I am also familiar with scripting languages like HTML, XML, XAML. & also have knowledge about web APIs (JSON data parsing etc.)

For testing and analyzing data, I can write scripts with Python, Matlab. I can also write small bash scripts for Linux.

I know many communication protocols like RS 422/485/232, TCP/IP, MilStd1553, Arinc429, UART, I2C, SPI, CANBus, GPIB, OneWire, Xaui, NMEA since I used them in many different projects.

I have high level domain knowledge of Tactical Data Link (Link16) & non-C2 systems, electro optical / thermal imaging systems (NUC, BPR, CLAHE), laser defense systems & INS/GPS systems. Also have knowledge about hardware test setups / ATEs. I am also very familiar with Nato Stanag documents & ICD documents.

I have graduate level electronics hardware design and debugging skills.

I am eager to learn new technologies and information. Therefore, I attended many on-site educations, and in my free time, I follow some online software courses over Udemy, Coursera, Youtube, Medium etc. platforms.

I have certificates about my C++ & C# skills.

EXPERIENCE

MilSOFT Yazılım Teknolojileri A.Ş.

Ankara, TR

Senior Software Engineer

April 2021 - Present

- SW Design and development in Link 16 Tactical Data Link System working on top of PowerPC embedded data link processor
- SW Development for "non-C2" systems running on top of VxWorks 653 RTOS
- Mil-Std-498 based software development, low level requirement development and unit testing
- writing SRS & SDD documents
- integration activities with Command & Control Systems
- working in SAFe environment

VxWorks653 RTOS | C/C++ | OOP | UML | Design Patterns | Scripting | SAFe

SKILLS

Programming: C, C++11, C# 6.0, Linux Bash, HTML, XML, MATLAB

Operating Systems: RTOS (VxWorks 653 & TI RTOS), Baremetal, Desktop(Windows & Linux), Embedded Linux, Android

IDEs: Visual Studio, Eclipse (Code Composer Studio, MCUXpresso, IAR Workbench), Qt, Labwindows CVI, Test Stand, Matlab, Android Studio

Developer Tools: Git with Gitlab / Bitbucket, Tortoise SVN, Jenkins, Jira, Confluence

Processors: TI C6000 Family (C6678), TI C2000 Family (Delfino & Piccolo), TI Tiva (Arm CortexM), TI MSP430, Microsemi SmartFusion2, NXP i.MX RT1050 (Arm CortexM7), PowerPC

Comm Protocols: RS232/422/485, Ethernet(TCP/IP, UDP), MilStd1553, Arinc429, Uart, I2C, SPI, CanBus, GPIB, OneWire, NMEA

Other Tools: Microsoft Office, Comport Tools(Teraterm, port monitor, comm operator), Wireshark, CoPilot

Domain Knowledge: Tactical Data Link Systems, Electrooptical imaging & targeting systems, Laser range finder & designator systems, INS/GPS systems

Aselsan MGEO Division

Ankara, TR

Senior Embedded Software Engineer

Sep 2018 - March 2021

- embedded software development for imaging & targeting gimbal systems (electro optical FLIR systems) which are mounted on top of UAVs
- embedded software running on top of TI RTOS with C++ for multicore digital signal processors
- designing & developing whole system management & communication software (integrating hardware components to software, defining & controlling RTOS tasks, controlling shared memory usage over DSP cores, implementing symbology & tracking software details, controlling of 3 different cameras, controlling all the logic, all of the communication of these units, and controlling of some other sub modules, logging system internals...)
- testing embedded software with debugging tools
- simulator / GUI software development with QT (C++) or Visual Studio (Visual C++ / C#)
- sripting, documenting, maintaining & researching

TI RTOS | C/C++ | C# | OOP | UML | Design Patterns | Linux | QT | Bash | Baremetal | Agile

Senior Embedded Software Engineer

Sep 2016 - Sep 2018

- embedded software development for hand-held laser range finder & target locator / designator systems
- baremetal embedded software & also driver level BSP with C/C++ languages
- designing & developing whole system management & communication software (integrating hardware components to software, controlling the current of laser diodes which are located on the part of laser transmitter, controlling & communicating with sub systems such as DMC (digital magnetic compass), military GPS module, integrated circuits, camera, receiver circuitry, temperature sensors, pressure sensors, motors, E2PROM etc.)
- testing embedded software with debugging tools
- simulator / GUI software development with Visual Studio (C# / WPF framework)
- documenting, maintaining & researching

Baremetal | C/C++ | C# | WPF | OOP | Design Patterns | Drivers | Agile

Systems Engineer & Software Developer in Test

July 2010 - Sep 2016

- user acceptance test (UAT) / environmental test (ET) software design & development to test the functionality of mass production units of INS/GPS & autopilot systems 1) manages custom test environment with many test hardwares like power supplies, oscilloscopes, load modules, digital multi meters, signal generators, burn-in chambers, rate tables, vibration platforms etc. & all of the communications of these hardwares over RS422/232 or ethernet 2) displays a responsive GUI for test operator to see status of all test environment hardware & messages of the system under test (over RS422/232, milstd1553, arinc429 or ethernet etc.) 3) verifies the functionality of a specific unit that is under test along with user acceptance test specifications 4) creates a detailed test log for all environmental or acceptance test details
- planning test scenarios after gathering client expectations & writing UAT/ET procedures
- building a custom automated test environments (ATE) consisting of test hardwares.
- analyzing test results, documenting, maintaining & researching

C | C# | LabwindowsCVI | TestStand | VisualStudio Forms | Matlab | ATE | Test Hardware

EDUCATION

Electrical & Electronics Engineering - BSc.(GPA 3.18/4.00)

Hacettepe University

Sep 2005 - Jun 2010

Specialised in Embedded Systems, Computer Architectures, Signal Processing and Control Systems. Me & two friends have designed a "Quadrotor" as graduation project. The main aim of the project was to design a Quadrotor with PIC18F4420 MCU, Sparkfun 6 axis freedom gyro, ESC, 4 motors (10000rev/min, 16A) propellers, and carbon fiber skeleton. We also had a camera under quadrotor to get images or videos. There was a Kalman filter implementation by software also. Every software component in order to make quadrotor fly in a stable way, was written by ourselves.

TRAININGS

Embedded Linux Training - Nazım Koç, Ankara, 2019
Software Architecture Design - Bedir Tekinerdoğan, Ankara, 2019
Intel OpenVino Workshop, Ankara, 2019
Matlab – Deep Learning Workshop, Ankara, 2019
C++ Programming for Embedded Systems - Doulos, Ankara, 2018
Advanced C# Programming Language - Salih Demiroğ, Ankara, 2018
Parallel Programming with NVIDIA CUDA, Ankara, 2018
System Calls in Linux - Nazım Koç, Ankara, 2017
C Programming for Embedded Systems - Doulos, Ankara, 2016
Labview Training - NI Turkey, Ankara, 2013

DIPLOMAS

Hacettepe University EEM - BSc.
Ayrancı Anatolian Highschool

LANGUAGES

Turkish	Native
English	Professional
Russian	Beginner

INTERESTS

Travelling	Gaming
Fitness	e-Sports
Skiing	Comics